Pilot 2 — Technical Protocol (Draft for External Review)

Working title: *Does an Agent's Own Change Keep Time? Testing an Experiential Index of Adaptation Against Clocks, Counts, and Logged Update Summaries*

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Status: draft protocol **v0.3**, 3 November 2026. Not frozen. To be frozen, hashed (SHA-256), and deposited on Zenodo before any main run. v0.2 revised v0.1 after an AI review (Google Argon) and an AI analysis of that review (Copilot); v0.3 revises v0.2 after an AI review of v0.2 (Google Deep Review, Argon), with clarifications after an AI review of v0.3 (Mistral Large 4). Every change and every rejected suggestion is listed in §19 and §20.
Builds on: Ionici (2026), *A Time of Their Own* v1.3, §4 and Level 2 of Table 2, https://doi.org/10.5281/zenodo.23847648; method of Ionici (2026), *Does Explicit Episode–Trace Binding Help?* v1.3, https://doi.org/10.5281/zenodo.23847602; code base https://doi.org/10.5281/zenodo.23847723.

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0. Position within the research programme (context for the reviewer)

This protocol is the empirical continuation of two published works by the same author. Neither is under review here. They are summarised so that every design decision below can be traced to its source.

0.1 The Perspective: what is to be tested

A Time of Their Own (v1.3) argues the following:

- **The problem.** A persistent artificial agent whose complete state can be saved and restored holds no evidence about elapsed time or about its own change that it could not have been given, reset, or forged together with that state.
- **The classification.** It classifies the temporal evidence available to an agent (order, duration, sequential computation, continuity, and its own change) by who can write, reset, forge, or copy it, and by whether the agent's self-model can read it.
- **Two roles that must be kept apart:**
    1. A **continuity anchor**: a non-restorable time reference c_a and a state-bound operation counter N, read through a trusted path. It detects and distinguishes suspension and restoration (Level 4 of its programme).
    2. A **dual clock**: an **experiential index** $c_e(t) = \Sigma_{\tau<t} D(S(\tau), S(\tau+1))$, accumulated change of the agent's persistent state S, read against c_a. It serves the agent's model of its own adaptation (Level 2). The index is non-decreasing, is updated online at constant cost, and is distinct from the net displacement $D(S(t_0), S(t_1))$. The choice of D is declared an open problem.
- **Admissibility of D.** A candidate D is admissible only if it:
    - (a) beats decoders restricted to within-episode (local) update statistics;
    - (b) survives the reset criterion and randomisation of episode order, where it is meant to reflect history;
    - (c) predicts a later functional consequence better than event counts and logged update summaries;
    - (d) is used by the agent rather than merely decodable from it.
- **Test programme.** Six levels (time reference, experiential index, history, discontinuity, closed-loop use, temporal self), each paired with the conventional competitor that can close it. A level is **closed** if its competitor is non-inferior on the level's primary metric within a margin fixed in advance and no worse on any declared cost dimension; a mixed trade-off is reported as unresolved, neither closing the level nor supporting the hypothesis. The Perspective states that each level needs its own evaluation contract; this

protocol is that contract for Level 2. The primary metric of Level 2 is **predictive validity per unit of storage**.
- **Stance on outcomes.** Equivalence with the competitor is a legitimate negative result, and *functional separation* (the platform suffices, and agent-held evidence is unnecessary) is a possible outcome of the programme. The Perspective offers its tests to others; versions 1.0–1.2 also stated that its author did not plan to pursue them. This study is a declared change of that plan.

This protocol tests Level 2 only: whether c_e, under some admissible D, is worth holding.

0.2 The closed study: how it is to be tested

Does Explicit Episode–Trace Binding Help? (v1.3, closed) asked whether binding an episode record to a summary of the internal change it produced (arm D) helps a model attribute the episode's source (own, imported, or inferred), compared with an information-matched control that receives the same streams without the binding (arm F). The model was a small Transformer self-model reading the traces of a plastic agent. Over five exploratory iterations:

- **The apparent advantage was a control failure.** Generic controls failed to use the trace, which produced a large but uninformative advantage for D (+0.38 Brier skill score). When the control was made competent (87% of a decoding ceiling), D − F was +0.002 BSS, with descriptive equivalence under a ±0.05 BSS margin.
- **The signal was local, not historical.** Decoder diagnostics showed that the source was carried by **local update statistics**, and that resetting the state before each episode did not reduce decodability. The benchmark therefore did not test attribution from path-dependent history.
- **A shortcut caused the apparent reversal.** With 32 episodes per window, an apparent reversal between objectives was traced to a shortcut in how one objective was applied, and removing it repaired the reversal.

What carries over is the method, not a result: an information-matched control, a competence audit of the control, a decoding ceiling, a reset criterion, equivalence tests (TOST) with a prespecified margin, the distinction between onset (whether learning starts) and skill (performance once learned), reliance versus necessity, frozen and hashed configurations, and a per-run inventory.

What carries over as a warning: local update statistics can impersonate accumulated history, and an under-specified control can manufacture an advantage. This is why competitor C6 is mandatory and why the baseline is deliberately strong (§7).

0.3 Why Level 2, and why not the continuity anchor

An earlier internal draft of Pilot 2 tested the continuity anchor (Level 4) in simulation. It was abandoned. With a simulated anchor that is deterministic by construction, classifying suspension against restoration reduces to comparing two counters. A raw anchor can then beat an information-matched adjudicated flag only where the flag is absent, which is an argument about availability, not an empirical finding.

Level 2 is different: whether c_e predicts the agent's own forgetting depends on the actual learning dynamics of the network, so the outcome is not fixed by the design.

Prior work with the same index. Feng et al. (2026, FOREVER, ACL 2026, arXiv:2601.03938) define a "model-centric time" as the cumulative L2 norm of parameter updates, $\tau_t = \Sigma_{i\leq t} \|\Theta_i - \Theta_{i-1}\|_2$ on LoRA weights. This is c_e^{D1} (their τ is reset at each new task). They use it to schedule replay at forgetting-curve intervals, with models from 0.6B to 13B parameters. Their only ablation of the time signal is against step counts; learning-rate-based and gradient-norm-based signals are not tested. Priority for the use of accumulated update

magnitude as a model clock for replay therefore belongs to FOREVER, and this study claims no novelty for the index itself.

What this study adds is an audit of identification: whether such an index carries information beyond the union of cheap online competitors that could reconstruct it (§7), under a placebo that destroys its temporal alignment (G4) and under an adversarial stream with much change and little interference (F6). If the answer is equivalence, the benefit attributed to a model-centric clock is reproducible from elementary optimizer statistics; if it is superiority, the clock carries information of its own. Both outcomes bear directly on that line of work. Forgetting prediction more generally also exists (§18).

0.4 Content, not provenance

In the Perspective, the difference between agent-held evidence and logged update summaries is partly one of **provenance**: who writes, holds, and can reset or forge the evidence. In a simulation every quantity is computed by the same code, so provenance cannot differ between predictors. **Stage A therefore tests content only**: whether the number c_e carries predictive information that the other cheap numbers do not. For parameter-space distances (D1, D1_rel, D2), c_e is itself a cumulative update summary; H1 then asks whether *this* summary beats the strongest set of *other* cheap summaries. Claims about provenance are outside the scope of this study.

0.5 Traceability of design decisions

Design decision	Source
Definition of c_e; distinction from displacement; D as open problem	Perspective §4
Q1–Q4 and the mapping to criteria (a)–(d)	Perspective §4 (admissibility)
Confirmatory baseline B_strong; descriptive B_standard	Perspective §5 (a level closes if the competitor matches); closed study (control competence); review v0.1 (§19)
Competitor C6 (local update statistics)	Closed study: main diagnostic finding
Primary metric: validity per unit of storage (H2)	Perspective §5, Level 2
Three-way outcomes (supported, equivalent, inconclusive) and TOST	Perspective (equivalence as a legitimate result); closed study (equivalence margins)
Ceiling G1, competence G3, onset versus skill G6, reliance G8	Closed study, method; Perspective §6
Frozen configurations, hashes, inventory, bit-identical reruns G7	Closed study and its code archive
Competitors C1, C5b–C5f, C8, C8b, C9	External literature on forgetting and consolidation (§18)
Stage C on pretrained language models	Perspective (claims concern persistent agents, not toy worlds)

0.6 What the reviewer is asked to evaluate

Only this protocol: whether its design can deliver an interpretable answer to Q1–Q4 at the stated scale. The claims of the Perspective and the results of the closed study are published and are not under review.

0.7 Glossary

Term	Meaning
S, θ	Persistent state of the agent; its parameters
m, v; $\hat{m}$, $\hat{v}$	Adam first and second moments; bias-corrected values

Term	Meaning
c_a	Elapsed time in ticks (§4.5)
N, N(t)	Committed optimizer steps; their count at time t
c_e^D	Experiential index: accumulated D-change of S
$\bar{\rho}$	Mean interval rate Δc_e / max(ΔN, 1); the only information in c_e beyond counts (§2)
Disp^D	Net displacement D(S(t_0), S(t_1))
√JS	Jensen–Shannon distance, base-2 logarithm, in [0, 1] (a metric)
m_ch	Chunk stride, in optimizer steps, for chunked distances (§5)
L_reh	Length of one rehearsal segment, 64 steps (§9)
q_0	Step quantum: 64 optimizer steps; segment lengths, idle intervals, the evaluation grid and the chunk strides are aligned to it (§4.3)
B_standard	Descriptive baseline: standard telemetry (§7)
B_strong	Confirmatory baseline: all cheap competitors (§7)
SESOI	Smallest effect size of interest
TOST	Two one-sided tests (equivalence)
MET	Minimal-effect test: lower confidence bound above the SESOI
BSS	Brier skill score (used in the closed study)
Gate	A prespecified validity check; failing it changes how a result may be reported

1. Research question

Let a Transformer learn continually from a non-stationary stream of tasks. Let

$$c_e(t) = \Sigma_{\tau<t} D(S(\tau), S(\tau+1))$$

be its *experiential index*: the accumulated change of its persistent state S under a distance D. It is a non-decreasing scalar held by the agent.

Storage. Maintaining c_e costs **one running scalar**. Answering a query about task k needs the snapshot c_e(t_{0,k}) taken at its last exposure. The marginal storage of adding c_e to any predictor set, and the total storage of the standalone trigger S-c_e, is therefore one running scalar plus **one scalar per tracked task exposure**: at most K + 1 scalars for K active tasks (here $K \leq 40$). The rate $\bar{\rho}$ (§2) needs nothing more, because N(t_{0,k}) is already stored for the baseline. For comparison, B_strong caches about 20 scalars per task exposure plus one global input counter (§7), and net displacement needs 0.03 M to 1.9 M numbers per task exposure (§5.2).

Q1 (predictive validity). Does Δc_e over an interval (t_0, t_1] predict the agent's forgetting of a task last trained at t_0 better than a strong set of cheap online competitors: elapsed time; step, token, and event counts; cumulative learning rate and loss; logged gradient summaries, including learning-rate-weighted ones; optimizer and parameter-state norms; local (within-window) update statistics; and the loss change on a fixed reference battery?

Q2 (storage efficiency). How much of the predictive value of the full net displacement D(S(t_0), S(t_1)), which requires storing a snapshot at t_0, does the scalar c_e retain?

Q3 (path dependence; secondary). Under a fixed multiset of experiences, do order-induced differences in forgetting track order-induced differences in Δc_e?

Q4 (use). Does a rehearsal scheduler that uses c_e outperform budget-matched schedulers that use the competitors, and does its advantage depend on the actual c_e stream rather than on a placebo stream?

The admissibility criteria for D stated in *A Time of Their Own* §4 map onto these questions:

Criterion	Question
(a) beats decoders restricted to local update statistics	Q1, competitor C6, gate G5
(b) order sensitivity where history matters	Q3 (secondary, §10.3)
(c) predicts a functional consequence better than counts and logged summaries	Q1
(d) used by the agent rather than merely decodable	Q4, gate G8

Scope restriction on criterion (b). The reset criterion is **not applicable** to the target used here: resetting $S(t_0)$ would erase the task whose forgetting is measured. Only order sensitivity is examined, and only as a secondary hypothesis. The rate of change $\rho(\tau)$ still depends on the whole prior history through $S(t_0)$; this is why absolute run progress $N(t_0)$ and state norms at t_0 are covariates (§7).

2. Identification note (read first)

Within a run, at a fixed evaluation time t_1, the ranking of tasks by $\Delta c_e = c_e(t_1) - c_e(t_\{0,k\})$ induces the same weak ordering as the committed steps $\Delta N_k = N(t_1) - N(t_\{0,k\})$ since their last exposure, with ties wherever the intervening steps produce no change ($D = 0$). Elapsed ticks Δc_a can differ where steps do not, for example across idle intervals. The reason is that c_e is non-decreasing. A global scalar index therefore carries **no task-specific ordinal information at a fixed time**.

It does carry **cardinal** information: the spacing between two tasks' values, $c_e(t_\{0,k\}) - c_e(t_\{0,j\})$, equals the change accumulated between their exposures, which varies with the rate $\rho(\tau) = dc_e/dN$. All information that c_e carries beyond counts lies in this **temporal variation of its rate**: some intervals of equal length contain more change than others. Three consequences follow:

1. Q1 is identifiable only to the extent that the interval rate $\bar{\rho} = \Delta c_e / \max(\Delta N, 1)$ varies across intervals and is not reconstructible from the baseline. Gate G2 checks this on the rate, not on the raw increment, because the raw increment is dominated by interval length. For the same reason c_e enters every predictor set as the pair $\{\Delta c_e, \bar{\rho}\}$ (§8), as the competitors do.
2. A scheduler that *selects which task to rehearse* by ranking tasks on Δc_e since their last exposure is equivalent to least-recently-rehearsed. Q4 must therefore test *when*, and *how much*, to rehearse (allocation over time), or use c_e together with task covariates inside a learned predictor (§9). Inside such a predictor the *conditional* ranking can differ from least-recently-rehearsed, because the spacing $c_e(t_\{0,k\}) - c_e(t_\{0,j\})$ is traded off against covariates at t_0; whether it does is checked on validation runs before H4a is interpreted (§9).
3. c_e is not task-specific memory. It cannot by itself encode which tasks a stretch of change interfered with (§10.3).

3. Agent

Component	Specification
Architecture	Decoder-only Transformer, 4 layers, d_model = 128, 4 heads, d_ff = 512, pre-LN, GELU, learned positional embeddings, maximum context 8 (examples have 4–6 tokens; a longer context would leave positional embeddings that never receive a gradient and only decay), vocabulary 512 tokens. Input embedding and output head **untied**; biases in all linear layers and LayerNorms. ≈ 0.93 M parameters (exact count recorded at Stage 0)
Sequences	One example per sequence, no packing; sequence length equals example length (4–6 tokens); padding to the batch maximum with masked loss
Optimizer (primary)	AdamW, β = (0.9, 0.999), ε = 1e-8, weight decay λ = 0.01, global gradient clipping at norm 1.0 applied immediately before `optimizer.step()`. m and v see the clipped gradient $\tilde{g}$. Weight decay applies to weight and embedding matrices, not to biases or LayerNorm parameters. m and v are initialised to 0 at the first childhood step, persist without reset through life, segment boundaries and warm restarts, and are bias-corrected with the global count of committed steps
Optimizer (robustness arm)	SGD with momentum 0.9, same clipping, learning rate and weight decay frozen at Stage 0. A passive EMA $v_\tau = 0.999\, v_{\tau-1} + 0.001\, \tilde{g}_\tau^2$ is maintained so that D2 is defined; it does not affect the update. Without preconditioning, D2 then scales with the square of the gradient magnitude (§5.1); the arm is exploratory
Persistent state S	Parameters θ and optimizer moments (m, v); distances are computed on θ
Logged per step	$\|g_\tau\|_2$ (raw) and $\|\tilde{g}_\tau\|_2$ (clipped); η_τ; batch size B_τ; training loss L_τ; $\|\Delta\theta_\tau\|_2$ before and after the decoupled weight-decay term; $\|\hat{m}_\tau\|_2$; $\lvert\langle \tilde{g}_\tau, \Delta\theta^{pre}_\tau\rangle\rvert$; $\|\theta_\tau\|_2$ and the norm of the decayed parameters; $\mathrm{Tr}\,\hat{v}_\tau$
Precision, device, determinism	float32 on CPU; `torch.use_deterministic_algorithms(True)`; fixed seeds for data, initialisation, and dropout (dropout = 0 in the primary arm). Each run is a **single-threaded worker** (`torch.set_num_threads(1)`, `torch.set_num_interop_threads(1)`, `OMP_NUM_THREADS=1`, `MKL_NUM_THREADS=1`); eight workers run in parallel
Software	Python 3.14.6, PyTorch 2.13.0 (CPU build), NumPy 2.5.1, scikit-learn 1.9.0, as installed at drafting; exact versions are re-recorded in `environment.txt` at freeze

4. World and streams

4.1 Task types and vocabulary

- **Fact tasks.** Task k is a set of 256 key→value pairs. A key is an ordered pair of key tokens, so the input is `[TASK_k] [K1] [K2] →` and the target is a single `[VALUE]` token. Keys are drawn from the target key vocabulary V_{target}, and the overlap between tasks is controlled by a pool-sharing parameter $\pi_{overlap}$. Values are drawn uniformly from 64 value tokens. Fact probes are **trained items**: they measure in-sample associative retention.
- **Rule tasks.** Task k is an algorithmic map $y = (a_k \cdot x + b_k) \bmod m_k$, with $m_k \le 64$ so that y is a single value token. The input is `[TASK_k] [D1] [D2] →`, where $x \in [0, 1023]$ is written as two base-32 digit tokens. Each rule task has a fixed training set of 256 inputs drawn from the 768 inputs that are not probes, and 256 held-out probe inputs: rule probes measure held-out algorithmic generalisation.
- **Reference tasks.** A fixed set of fact and rule tasks, trained only during the childhood phase. They are never targets. They supply the reference battery R (256 items: 128 fact, 128 rule) and the behavioural battery B_{beh} (256 items: 128 fact, 128 rule, disjoint from R). Their keys use the key vocabulary V_{ref}, and their rule moduli are not used by any target task.
- **Distractor tasks (F6).** Fact tasks on the distractor key vocabulary $V_{distractor}$, with their own task tokens.
- **Vocabulary partition** (frozen at Stage 0): 64 task tokens (40 target, 8 reference, 8 distractor, 8 spare); key tokens in three **token-disjoint** vocabularies, V_{target} (180 tokens), V_{ref} (84) and $V_{distractor}$ (84);

64 value tokens; 32 digit tokens; at most 4 special tokens (→, padding, spares); 512 in total. Disjointness holds for single tokens, not only for key pairs.

- **Exposure.** Every training, re-exposure, and rehearsal segment iterates over balanced permutations of the task's items (full epochs). With 256 items and batch sizes 16, 32 or 64, an epoch takes 16, 8 or 4 steps; every segment length is a multiple of 64 steps (§4.3), so every segment ends at the end of an epoch and every item of halves A and B has the same exposure count.

4.2 Probes

Each target task has a probe set: its 256 facts (fact tasks) or 256 held-out inputs (rule tasks). The probe set is split **once, at task creation**, into disjoint halves A and B of 128 items each.

- **Half A** supplies the confirmatory target T1.
- **Half B** supplies the competence covariates at t_0, the inclusion criterion, the reliability ceiling (with A), and the Stage B retention target T3.

No probe item belongs to R or to B_beh. All probe losses and probabilities are read at the single answer position after →; cross-entropy is in nats.

4.3 Run structure

Phase	Steps	Content
Childhood	exactly 3,072 (48 × 64)	Reference tasks. Gate: reference accuracy ≥ 0.95 at the end; a run that fails is replaced and counted
Life	≈ 20,000 (a multiple of 64)	≈ 40 target tasks, presented according to the stream family; rehearsal segments, re-exposure segments, and idle intervals according to the family
Evaluation	—	Probes of every learned task are evaluated at the end of every segment, at the end of every idle interval, and on a grid every 256 optimizer steps (during idle intervals, every 256 ticks)

Step quantum. All segment lengths, idle intervals, warm-restart times and the evaluation grid are integer multiples of $q_0 = 64$ optimizer steps (ticks, for idle intervals). Every chunk stride m_ch considered in §5.1 divides 64, so segment boundaries and evaluation points always coincide with chunk boundaries. Chunks are indexed by committed optimizer steps from the first step of life.

Segment boundaries. A segment boundary is an agent-observable event: a change of the task token in the training inputs, the start or end of an agent-triggered rehearsal, or a change of batch size. Warm restarts in F3 change only the learning rate and are not segment boundaries. No generator metadata defines a boundary.

4.4 Stream families

Family	Generator	Intended decoupling	Role
F1, natural	New task every fixed number of steps, fixed segment length, round-robin rehearsal, constant LR, fixed batch	None	Positive control, reported alone: counts are expected to suffice; **not pooled** into H1
F2, variable novelty	New tasks interleaved with re-exposure to already-mastered data in random amounts; batch size ∈ {16, 32, 64} per segment	Steps and tokens decouple from change	**H1 confirmatory domain** (pooled with F3)
F3, LR schedule	Cosine annealing with warm restarts at random times (multiples of 64 steps)	Steps decouple from change; LR-weighted summaries are strong competitors	**H1 confirmatory domain** (pooled with F2)

Family	Generator	Intended decoupling	Role
F4, idle	Random idle intervals with no updates; c_a advances, θ is frozen	Only elapsed time decouples (C1–C7 stop with c_e)	Sanity check only; **not pooled** into H1
F5, held-out	Mixture of the F2–F3 generators with a different proportion of rule tasks, a different π_overlap, and idle intervals	Generalisation across generator parameters	Never used for fitting or selection; test only; replication check reported alone (§10.2)
F6, adversarial	Target exposures at a regular rate, interleaved with variable-length distractor segments on V_distractor with their own task tokens, label noise 0.3, and batch size ∈ {16, 32, 64} per segment as in F2, under the same optimizer configuration	High parameter change with little interference: favours the competitors	Never used for fitting or selection; test only; adversarial check reported alone (§10.2)
O, order blocks	Designed blocks (§10.3) forked from one saved state	Path dependence	H3 only; **not pooled** into H1, H2, H4

F5 is built from the F2–F3 generators, so it tests generalisation across generator parameters, not freedom from designed decoupling. F6 tests the converse regime that F2–F5 lack: much movement, little forgetting. Scale-drift interventions (weight-decay pulses, rescaling of pre-LN weights) are exploratory robustness checks only, because they have no counterpart in ordinary training. Generator parameters for each family are frozen with the protocol.

4.5 Time

c_a counts *ticks*. Each optimizer step adds 1 tick; each idle interval adds its length in ticks; evaluations add 0 ticks. Idle intervals are inserted only between segments, in multiples of 64 ticks. Without idle intervals c_a equals N. All interval sums in §7 run over committed optimizer steps $\tau \in (N(t_0), N(t_1)]$, so idle ticks contribute nothing to them.

5. Distances D and the index c_e

5.1 Candidate distances

$\Delta\theta^{pre}(\tau)$ denotes the update of step τ before the decoupled weight-decay term.

D	Definition	Evaluated
D1	$\|\theta(\tau+1) - \theta(\tau)\|_2$, the full update including weight decay	Every step
D1_rel	$\left(\sum_\ell \|\Delta\theta^{pre}_\ell(\tau)\|_2^2 \,/\, (\|\theta_\ell(\tau)\|_2^2 + 10^{-12}) \right)^{1/2}$, where ℓ runs over the weight matrices and embedding matrices (biases and LayerNorm vectors excluded)	Every step
D2	$\left(\sum_i \hat{v}_i(\tau - L)\, (\Delta\theta^{pre}_i(\tau))^2 \right)^{1/2}$, with $\hat{v}$ the bias-corrected second moment lagged by L = 256 steps	Every step
D3	Mean over x ∈ R of $\sqrt{JS}(p_{S(\tau)}(\cdot\|x), p_{S(\tau+m_{ch})}(\cdot\|x))$ at the answer position, |R| = 256	Every m_ch steps (chunked)
D4 (exploratory)	√(1 − CKA), with standard (biased) centred linear CKA of the final-layer answer-position representations on R, taken after the final LayerNorm (Kornblith et al. 2019)	Every m_ch steps (chunked)

Why D2 uses a lagged curvature. Under Adam, $\Delta\theta^{pre}_i \approx -\eta\, \hat{m}_i / (\sqrt{\hat{v}_i} + \varepsilon)$. Weighting by the *current* $\hat{v}$ therefore cancels the preconditioner: $(\sum_i \hat{v}_i(\tau)\, \Delta\theta^{pre}_i(\tau)^2)^{1/2} \approx \eta_\tau \|\hat{m}_\tau\|_2$, so a same-step D2 would be the learning-rate-weighted momentum norm, a logged optimizer summary (C5d, §7). A fixed lag avoids exact

cancellation without using segment boundaries, which an agent in an unsegmented stream would not have. With $\beta_2 = 0.999$ the lagged and current moments stay close over 256 steps, so D2 may remain nearly collinear with C5d; gate G2 detects this, and C5d stays in the baseline. Under SGD (robustness arm) the update is not preconditioned, so D2 scales with the square of the gradient magnitude; the difference is reported, not interpreted.

Why D1_rel. In a pre-LN Transformer, some weight rescalings change $\|\theta\|$ without changing the function, and AdamW shrinks θ at every step independently of the data. D1_rel removes both scale effects. Raw D1 includes the weight-decay step and is kept for comparison; the corresponding weight-decay path is a competitor (C5f, §7).

- c_e^D(t) = sum over steps (D1, D1_rel, D2) or over chunks (D3, D4) up to t.
- √JS is a metric, so the chunked D3 sum is a lower bound on the per-step sum. √(1 − CKA) is the chordal distance between centred Gram matrices normalised to unit Frobenius norm and also satisfies the triangle inequality.
- **Chunking control.** On calibration runs, chunked versions of D1, D1_rel and D2 at the same stride m_ch are computed alongside the per-step versions, so that a difference between D3 and D1/D2 cannot be produced by temporal smoothing alone. A chunked distance over $[\tau, \tau + m_{ch})$ uses the vector sum of the updates in the chunk (pre-decay for D1_rel and D2), weighted with the layer norms at τ (D1_rel) or with $\hat{v}(\tau - L)$ (D2). Sensitivity: $m_{ch} \in \{1, 4, 8, 16, 32, 64\}$ on 5 calibration runs ($m_{ch} = 1$ over a 2,048-step window). $m_{ch} = 8$ is adopted if, for D3, the within-run Spearman correlation between Δc_e at $m_{ch} = 1$ and $m_{ch} = 8$ across intervals is ≥ 0.98 and their ΔR^2 differ by ≤ 0.01; otherwise $m_{ch} = 4$ (with $|R|$ reduced to 128 if needed for cost).
- **Stationarity check of R.** Because R is never retrained after childhood, the sensitivity of D3 may decay as R is forgotten. On F1 calibration runs, the mean D3 rate in the fourth quartile of the life phase must be within ±20% of the first quartile. If not, the reference tasks are split into R_maint, rehearsed with one batch every 512 steps in all families, and R_eval, never trained after childhood; D3, D4, C10 and C10b then use R_eval only, and the calibration runs are repeated before freeze.
- **Moment-reset diagnostic (calibration only).** On 4 calibration runs, a forked copy resets the optimizer moments at every segment boundary; the change in the per-step rate of each candidate D is reported, to show how much of ρ is carried by optimizer state rather than by the parameters.
- **D*** is the single D used in the confirmatory tests. It is chosen among D1, D1_rel, D2 and D3 on calibration runs, out of fold (§12a), as the one with the largest incremental validity over B_strong (§7), and is frozen before any test run.

5.2 Net displacement and its storage

$\mathrm{Disp}^D(t_0, t_1) = D(S(t_0), S(t_1))$, evaluated at grid points: $\|\theta(t_1) - \theta(t_0)\|_2$ for D1; the layer-wise relative form with the layer norms at t_0 for D1_rel; $(\Sigma_i \hat{v}_i(t_0)(\theta_i(t_1) - \theta_i(t_0))^2)^{1/2}$ for D2; and the D3 and D4 distances between the cached outputs at t_0 and the outputs at t_1. The cached object per active task exposure is:

D	Cached at t_0	Numbers per t_0
D1, D1_rel	$\theta(t_0)$ (and layer norms)	≈ 0.93 M
D2	$\theta(t_0)$ and $\hat{v}(t_0)$ (Fisher weight anchored at t_0, as in EWC)	≈ 1.9 M
D3	answer-position output distributions on R	256 × 512 ≈ 0.13 M
D4	final-layer answer-position representations on R	256 × 128 ≈ 0.03 M

A snapshot of task k is overwritten at its next exposure, so at most $K \leq 40$ snapshots are active. Storage ratios in H2 are reported per D (of order 10^{-5} to 10^{-6} in Stage A). The path metric of D2 is lagged while its displacement is anchored at $\hat{v}(t_0)$; if D* = D2, the metric-free Disp^{D1} is reported alongside, so that a difference between

path and displacement is not confused with the choice of curvature anchor. On test runs, snapshots are kept only for D*; C8b and C9 are evaluated on a subsample of 30 test runs (§14).

6. Targets

T1 (primary): forgetting. For each target task k and each exposure segment of k (training, re-exposure, or rehearsal) ending at step s_0, let t_0 be the evaluation immediately after s_0. For every evaluation time $t_1 > t_0$ strictly before the first step of the next segment that trains k:

$F_k(t_0, t_1) = \bar{L}_{A,k}(t_1) - \bar{L}_{A,k}(t_0)$,

where $\bar{L}_{A,k}$ is the mean cross-entropy on probe half A of task k.

- **Inclusion:** a triplet is included only if the probe-**B** accuracy of task k at t_0 is ≥ 0.90, and either $\Delta N = N(t_1) - N(t_0) \geq 64$, or $\Delta N = 0$ and $\Delta c_a \geq 64$ ticks (idle intervals). Excluded triplets are counted and reported by reason.
- **Unit of observation:** the triplet (k, t_0, t_1). **Unit of inference:** the run.
- **Weighting:** each exposure episode (k, t_0) carries equal total weight within its run: $w = 1 / (E_r \cdot M_{r,k,t_0})$, where E_r is the number of valid episodes in run r and M_{r,k,t_0} the number of evaluation times of that episode. The same weights are used in model fitting (`sample_weight`) and in R^2_r.
- **Strata (descriptive).** The H1 statistic is also reported by task type (fact: in-sample retention; rule: held-out generalisation) and by tercile of ΔN.

T1-acc (secondary): bounded forgetting. $\Delta Acc_{A,k} = Acc_{A,k}(t_0) - Acc_{A,k}(t_1)$ on the same triplets. It checks that an effect on T1 reflects functional forgetting rather than logit-scale drift. The H1 statistic is reported for T1-acc descriptively.

T2 (secondary): behavioural drift. The √JS distance between the outputs of $S(t_0)$ and $S(t_1)$ on B_{beh}. It is reported descriptively. T2 is structurally favourable to $Disp^{D3}$, so it is not used for confirmatory claims.

T3 (Stage B): retention. The time-averaged probe-B accuracy over all learned tasks during the life phase, at a fixed rehearsal budget.

7. Predictor sets and competitors

All interval quantities are sums over $\tau \in (N(t_0), N(t_1)]$. Every competitor is agent-available online and costs ≤ 2 stored scalars per t_0. $\Delta\theta_\tau$ means the update before the decoupled weight-decay term ($\Delta\theta^{pre}_\tau$) unless stated otherwise.

Covariates X_cov, given to every model (agent-observable only):

- task type (fact or rule), and the log of the number of distinct target values in the task's training set (64 for fact tasks, m_k for rule tasks), which fixes the scale of the loss;
- $\bar{L}_{B,k}(t_0)$ and $Acc_{B,k}(t_0)$ (competence at t_0, from half B);
- $\bar{L}_{B,k}$ at the start of the exposure segment ending at s_0 (prior mastery, a proxy for novelty);
- input overlap of task k with the agent's history: $1 - \sqrt{JS}(q_k, q_{hist}(t_0))$, where q_k is the distribution of the task's input **pairs** (key pairs, or digit pairs for rule tasks; task tokens and → excluded) and $q_{hist}(t_0)$ the cumulative distribution of input pairs from the first step of life up to t_0 (one counter over input pairs);
- $N(t_0)$, absolute run progress; task tenure $N(t_0) - N(t_{first,k})$, with $t_{first,k}$ the evaluation after task k's first training segment; number of prior exposures of k; number of active learned tasks $K_{active}(t_0)$;
- $\|\theta(t_0)\|_2$ and $\sqrt{Tr}\,\hat{v}(t_0)$.

The generator parameter $\pi_{overlap}$ is **not** a covariate: it is world metadata (see C8).

B_standard (standard telemetry; descriptive only):

Code	Predictor over $(t_0, t_1]$	Source
C0	Δc_a, elapsed ticks	wall clock
C1	ΔN, with the fixed basis $\{\Delta N, \log(\Delta N + 1), \log(\Delta N + 10), \log(\Delta N + 100)\}$	shifted power law of Kalajdzievski (2024), arXiv:2401.05605
C2	Δ tokens	data volume
C3	$\Sigma \eta_\tau$	cumulative learning rate
C4	ΣL_τ	cumulative training loss
C5	$\Sigma \|\|g_\tau\|\|_2$ (raw)	logged gradient norm
C7a	number of segment boundaries (§4.3) and of agent-triggered rehearsals	agent-observable event counts

B_strong (confirmatory baseline) = X_cov ∪ B_standard ∪:

Code	Predictor over $(t_0, t_1]$	Source
C5a	$\Sigma \|\|\tilde{g}_\tau\|\|_2$ (clipped)	gradient actually used by the optimizer
C5b	$\Sigma \eta_\tau \|\|g_\tau\|\|_2$, $\Sigma \eta_\tau \|\|\tilde{g}_\tau\|\|_2$, $\Sigma \eta_\tau L_\tau$	within-step products that interval sums cannot recover
C5c	$\Sigma \eta_\tau^2 \|\|\tilde{g}_\tau\|\|_2^2$, $\Sigma \lvert\langle \tilde{g}_\tau, \Delta\theta_\tau \rangle\rvert$	squared-gradient accumulator; path-integral work of Synaptic Intelligence (Zenke et al. 2017)
C5d	$\Sigma \eta_\tau \|\|\hat{m}_\tau\|\|_2$	learning-rate-weighted momentum norm (the same-step D2 identity, §5.1)
C5e	$\Sigma \|\|\hat{m}_\tau\|\|_2$	unweighted momentum norm
C5f	$\Sigma \eta_\tau \lambda \|\|\theta^{dec}_\tau\|\|_2$, with θ^{dec} the decayed parameters	path length of the decoupled weight-decay term
C2b	$\Sigma \eta_\tau / B_\tau$	noise-scale-weighted steps
C6	mean and SD (population, ddof = 0) of $\|\|\Delta\theta_\tau\|\|_2$ over the first W_eff and the last W_eff steps of $(t_0, t_1]$, with $W_eff = \min(16, \lfloor \Delta N/4 \rfloor)$, windows clamped inside the interval; computed for the pre-decay and for the post-decay update; all set to 0 when $\Delta N = 0$	local update statistics (closed study; criterion (a))
C6b	$\|\|\theta(t_1)\|\|_2 - \|\|\theta(t_0)\|\|_2$ and $\sqrt{\mathrm{Tr}}\,\hat{v}(t_1) - \sqrt{\mathrm{Tr}}\,\hat{v}(t_0)$	state-scale drift (signed)
C10	$L_R(t_1) - L_R(t_0)$: change of the mean loss on the reference battery R	canary loss (signed); 1 stored scalar per t_0
C10b	Σ over chunks of $\lvert L_R(\tau + m_ch) - L_R(\tau)\rvert$, from the same forward passes as D3	accumulated unsigned canary change: path accumulation on R without the √JS distance
rates	$Z / \max(\Delta N, 1)$ for $Z \in \{$C2, C2b, C3, C4, C5, C5a, C5b, C5c, C5d, C5e, C5f, C10, C10b$\}$ (the rate of C10 is signed)	computed, not stored

Why two baselines. B_strong is deliberately close to a reconstruction of c_e from cheap telemetry, so that a positive H1 cannot be explained by an omitted product of learning rate and gradient, by momentum, by weight decay, by norm drift, by local statistics, or, for D3, by path accumulation of a cheap loss on R. Claimed

advantages over under-trained baselines often vanish when the baseline is trained properly (Mukhoti et al. 2018); the closed study documented the same failure for an information-matched control. B_standard is reported so that readers can see how much of any gap is closed by the specialised competitors. If c_e beats B_standard but not B_strong, the result is reported as *reconstructible from cheap telemetry*, not as support.

B_strong^{+} (exploratory sensitivity). Further norm- and preconditioner-scaled accumulators ($\Sigma\, \eta\|\hat{m}\| / \|\theta\|$, $\Sigma\, \eta\|\hat{m}\| / \sqrt{(\mathrm{Tr}\, \hat{v} / P)}$, and the ratio of lagged to current $\mathrm{Tr}\, \hat{v}$) are reported as a sensitivity baseline, not in the confirmatory comparison.

Reference predictors (not agent-available, or of high storage; used to bound, not to compete; exploratory):

Code	Predictor	Storage
C7b	separate counts of new-task and re-exposure segments (oracle labels)	world metadata
C8	Σ over intervening tasks j of key overlap(k, j) × steps on j; for rule tasks, overlap is the Jaccard overlap of input pairs	world metadata (Lee, Goldt and Saxe 2021, arXiv:2107.04384)
C8b	$\mathrm{Disp}^{D2}(t_0, t_1)$ with $\hat{v}(t_0)$, Fisher-weighted displacement (EWC: Kirkpatrick et al. 2017)	≈ 1.9 M per t_0
C9	First-order response $\langle \nabla_\theta L_{\mathrm{train},k}(S(t_0)), \theta(t_1) - \theta(t_0) \rangle$, with the gradient computed on task k's **training items only**, never on probe half A (cf. Su and Zhang 2026; Jin and Ren 2024)	≈ 1.9 M per task
C8b + C9	Second-order Taylor reference, from the same cached objects; reported alongside each alone	≈ 2.8 M per task
Ceiling	Reliability of the half-A target: within-run, episode-weighted correlation r_{AB} between F_A and F_B on the same triplets (weights of §6), **uncorrected** (the target uses one half)	—

8. Prediction models and competence audit

- **Primary model family:** gradient-boosted regression trees (scikit-learn `HistGradientBoostingRegressor`), `max_features = 1.0`, `max_bins = 255`. Monotone constraints (`monotonic_cst`, applied to every column of a multi-column code): +1 for C0, all four terms of C1, C2, C2b, C3, C4, C5, C5a–C5f, C10b, Δc_e, Disp, and, in exploratory reference models, C8 and C8b; 0 for C7a, C6, C6b, C10, all rates (including $\bar{\rho}$ and the rate of Disp), all of X_cov, and, in exploratory models, C7b and C9. An unconstrained fit (`monotonic_cst = None`) is reported as a sensitivity check.
- **How c_e and Disp enter.** Adding c_e to a predictor set adds the pair $\{\Delta c_e^{D^*}, \bar{\rho}^{D^*} = \Delta c_e^{D^*} / \max(\Delta N, 1)\}$, at no extra storage; Disp^{D^*} enters the same way. c_e thus has the same representation as every competitor in B_strong, and H1 tests the quantity that G2 audits. Because ΔN is in B_strong, $\bar{\rho}$ is a deterministic function of Δc_e and ΔN, so the pair adds one degree of freedom; the rate is given explicitly because trees cannot form ratios, and G2 audits exactly that degree of freedom.
- **Search space** (the same for every predictor set): `learning_rate` ~ LogUniform[0.01, 0.2]; `max_iter` ∈ {100, 200, 400}; `max_leaf_nodes` ∈ {15, 31, 63}; `min_samples_leaf` ∈ {20, 50, 100}; `l2_regularization` ~ LogUniform[10^{-4}, 10].
- **Secondary model family:** ridge regression with log1p applied to raw positive interval sums (including ΔN and Δc_e), the same shifted-log basis for Δc_e as for ΔN (knots scaled by the mean training-run increment per step), the rates as linear terms, $\mathrm{sign}(x) \cdot \log(1 + |x|)$ for signed predictors (C6b, C10, C9), pairwise interactions between each interval predictor and each variable of X_cov, a `StandardScaler` fitted on training runs, and λ chosen by cross-validation grouped by run.
- **Equal tuning budget:** every predictor set receives the same random search, 50 configurations, on validation runs, with the same seeds. The best configuration of B_strong is always included as a

candidate for B_strong + c_e. The **locked-hyperparameter** ΔR^2 (both sets fitted with the B_strong configuration) is the confirmatory statistic; the re-tuned ΔR^2 is reported alongside. **Divergence rule:** if the locked and the re-tuned ΔR^2 lead to different outcomes and differ by more than 0.01, H1 is reported as *inconclusive (sensitive to hyperparameters)*, never as closing Level 2.

- **Fitting:** models are fitted on pooled, weighted triplets from the training runs and evaluated on test runs. No test-run information is used in fitting, selection, or transformation. **Training-sample variance:** for both model families, 250 bootstrap resamples of the training runs (with replacement; refitted with locked hyperparameters; any cross-validation grouped by original run) are evaluated on the test runs, and the confirmatory interval resamples training replicates and test runs jointly (§10.1). Quantities on calibration runs are computed out of fold (§12a).
- **Skill per test run r:** weighted $R^2_r = 1 - SSE_r / SST_r$, with SST_r around the weighted mean of run r. Weighted Spearman ρ_S,r and MAE_r are reported alongside.
- **Decomposition (reported, not confirmatory):** within-episode skill, with SST computed around the mean of each episode (k, t_0) across its t_1; and partial skill [SSE_r(B_strong) − SSE_r(B_strong + c_e)] / SSE_r(X_cov).
- **Stage B deployment model.** The predictors used by S-pred in Stage B are fitted separately on the same Stage A training runs, with the competence covariates computed as Stage B computes them online (loss and accuracy over the last epoch of the segment, and over its first epoch for prior mastery), and without the inclusion filter. Their in-loop calibration (MAE and ρ_S against half-A probes) is logged during Stage B.

9. Stage B: use by the agent

Stage B is run only if H1 is supported (§10.1).

- **Budget.** The rehearsal budget is fixed per run: K_reh = Q / L_reh rehearsal segments of L_reh = 64 steps over the life phase. The budget is a **hard cap**, and every scheduler spends exactly Q on every run (asserted per run; gate G7).
- **Decision grid.** Every scheduler, including S-pred and S-oracle, decides only on the 64-step grid; S-pred scores all active tasks in one batched call per decision.
- **Pacing.** A threshold scheduler triggers a rehearsal when its statistic since the last rehearsal reaches $\theta_eff(N) = \theta_cal \cdot [(N_life - N) / N_life] / \max(\varepsilon, (K_reh - k_spent(N)) / K_reh)$, with $\varepsilon = 1 / (2\ K_reh)$. Spending ahead of schedule raises the threshold; falling behind lowers it. When the remaining life steps fall to the remaining budget, all remaining rehearsals are executed (terminal flush). θ_cal is calibrated separately for each threshold scheduler on validation runs, to minimise the mean absolute deviation between cumulative spending k_spent(N)/K_reh and uniform spending N/N_life; the terminal flush must average ≤ 5% of K_reh on validation runs, and its size is logged per run (G7).
- **Schedulers.** By the identification note (§2), schedulers differ in **when** they spend the budget, and in which task only through a learned predictor that includes covariates.

Scheduler	Rule
S-uniform	One rehearsal segment every ⌊N_life / K_reh⌋ optimizer steps (rounded to the grid); task chosen least recently rehearsed
S-c_e	Rehearsal triggered when Δc_e^{D*} since the last rehearsal ≥ θ_eff; task least recently rehearsed
S-C2, S-C3, S-C4, S-C5, S-C5b, S-C7a	As S-c_e, with the trigger on tokens, cumulative LR, cumulative loss, gradient norm, LR-weighted clipped gradient norm, or event counts
S-pred(B_strong)	The Stage B deployment model on B_strong (§8) scores every task; a rehearsal is triggered when the increase in total predicted forgetting over active tasks since the last rehearsal reaches θ_eff, and the task with the largest predicted forgetting among tasks with ΔN ≥ 64 is rehearsed

Scheduler	Rule
S-pred(B_strong + c_e)	Same, with {Δc_e^{D*}, ρ̄^{D*}} added (information-matched pair)
S-oracle	Rehearse the task with the largest true current forgetting on half B, evaluated on the decision grid (upper bound)
S-random	Random times and tasks, same budget

- **Relation to FOREVER.** With D* = D1, S-c_e is a trigger on the clock of Feng et al. (2026). An exploratory arm S-FOREVER applies their schedule (replay when c_e^{D1} since the end of a task's training reaches d · τ_day, with τ_day the change over the first 24 steps and d ∈ {1, 2, 4, 7, 15, 30, …}), under the same budget cap and pacing.
- **Comparator selection.** The single best trigger competitor S-C* is chosen among S-uniform, S-C2, S-C3, S-C4, S-C5, S-C5b and S-C7a on **validation runs**, frozen, and evaluated once on test runs.
- **Task-selection check.** On validation runs, S-pred(B_strong) and S-pred(B_strong + c_e) must select a task other than the least recently rehearsed one on at least 10% of rehearsals. Otherwise H4a is reported as a duplicate of the timing test H4b, not as a test of task selection.
- **Online covariates.** In Stage B, competence covariates at t_0 are computed online from the training batches, as in the deployment model (§8). T3 uses half B only, and no scheduler except S-oracle reads any probe.
- **Placebo for reliance (G8).** Information that is decodable need not be used (Elazar et al. 2021); reliance is therefore tested by intervention. For each Stage B test seed, the live run of S-c_e (or of S-pred(B_strong + c_e)) is executed first. Its realised per-step c_e increments, cut into 256-step blocks with idle zeros kept in place, are permuted with a fixed seed and fed online, step by step, to a placebo twin with the same seed. For S-pred, the permuted quantity is the out-of-fold residual of Δc_e given B_strong, so that the feature pairs stay within the range seen in training. A cross-seed yoked stream from the same family, aligned by N, is run as a robustness check.
- **Logged:** realised rehearsal steps and terminal-flush size per run; Wasserstein distance between the trigger times of S-c_e and of S-C* (when); task agreement at matched times (which); in-loop calibration of S-pred.

10. Hypotheses

10.1 Decision rules

- **One-sided level:** α = 0.025 per side for every test. Each hypothesis is decided from a single two-sided (1 − 2α_adj) BCa bootstrap confidence interval (10,000 resamples over the resampling unit; for Stage A statistics, each draw resamples the test runs within family strata and pairs them with one of the 250 training-run replicates of §8, drawn uniformly; the BCa acceleration is estimated by the jackknife over test runs), where α_adj is the level assigned to it by the procedure below:
  - **supported** if the lower bound is > δ (minimal-effect test);
  - **equivalent** if the whole interval lies within (−δ, δ) (TOST, each side at α_adj; Lakens et al. 2018; for stochastic machine-learning experiments, Knopov and Shvai 2026);
  - **inconclusive** otherwise. H2 uses the non-inferiority rule of §10.2, because its margin is built into its contrast.
- **Multiplicity (fixed sequence, then Holm).** H1 is tested first at α = 0.025. Only if H1 is supported are H2, H4a and H4b tested, as a Holm family at α = 0.025: ranked by their one-sided bootstrap p-values, the j-th is tested at α_adj = 0.025 / (3 − j + 1). This controls the family-wise error rate. If H1 is equivalent or inconclusive, H2–H4 are not tested confirmatorily and Stage B is not run. The family checks of §10.2 (F1, F5, F6) are interpretation conditions, not members of the Holm family; each uses one-sided α = 0.025.

- **SESOI (rule frozen now, value computed on calibration runs and frozen before test):** $\delta = \max(0.02, \min(0.05, 0.15 \times [\text{ceiling} - \bar{R}^2_cal(B_strong)]))$, where the ceiling $\bar{r}_AB$ (G1) and $\bar{R}^2_cal(B_strong)$ are averaged over the F2–F3 calibration runs, out of fold (§12a). Raw ΔR^2 and ΔR^2 as a fraction of the headroom are both reported. Stage B: δ_B = 2 percentage points of T3.

10.2 Confirmatory hypotheses and family checks

	Statement	Statistic	Resampling unit
H1	Adding c_e to B_strong increases held-out skill on T1	Mean over the H1 test runs (**F2 and F3**, pooled) of the locked-hyperparameter $\Delta R^2_r = R^2_r(B_strong + c_e) - R^2_r(B_strong)$, where "+ c_e" adds $\{\Delta c_e^{D*}, \bar{\rho}^{D*}\}$	test run, stratified by family
F5 replication	The H1 effect generalises across generator parameters	The same statistic on F5 test runs; **passed** if the lower bound is > 0	F5 test run
F6 adversarial check	c_e is not misleading under change without interference	The same statistic on F6 test runs; c_e is **misleading under drift** if the upper bound is < 0	F6 test run
F1 positive control	Counts suffice in the natural stream	The same statistic on F1 test runs, reported alone; equivalence is the expected result	F1 test run
H2	The scalar index retains at least 80% of the displacement's incremental value	$\bar{Z}$ = mean over the H1 test runs of $Z_r = \Delta R^2_r(c_e) - 0.80 \cdot \Delta R^2_r(Disp^{D*})$. **Supported** (retains ≥ 80%) if the lower bound of $\bar{Z}$ is ≥ 0; **refuted** (retains < 80%) if the upper bound is < 0; inconclusive otherwise. Tested only if G2b passes. The ratio of means and its Fieller interval are reported descriptively	test run, stratified by family
H4a	c_e helps a covariate-aware rehearsal predictor	Mean paired difference in T3, S-pred(B_strong + c_e) − S-pred(B_strong); SESOI δ_B	matched Stage B seed pair
H4b	c_e suffices as a standalone rehearsal trigger	Mean paired difference in T3, S-c_e − S-C*; SESOI δ_B	matched Stage B seed pair

10.3 Optional exploratory module H3 (path dependence)

H3 is **not** part of the confirmatory study. Its design is preregistered here so that it cannot be tuned after the fact, but it is run only if H1 is supported and the compute budget of §12a remains after Stages A and B (it needs about 120 runs, comparable to the whole Stage A test set). If run, it is reported as exploratory.

The v0.1 design (segments X and Y with high and low key overlap with k) is withdrawn: a global index cannot encode *which* task a stretch of change interfered with, and recency can reverse the sign of the order effect. The redesign varies a **global** property:

- **Blocks.** From one saved state $S(t_0)$ after task k is learned (θ, m, v, data-generator state), two branches are forked. There are 30 saved states, each forked under 2 independent mini-batch permutation seeds (60 matched pairs). Segment A trains a new task with the same input-token overlap with k as segment B; segment B re-exposes already-mastered data. A two-stage learning-rate schedule (η_high, then η_low) is fixed. Branch 1: A at η_high, then B at η_low. Branch 2: B at η_high, then A at η_low. Steps, tokens, batches, the multiset of examples, $\Sigma\ \eta$, and C7a are identical across branches.
- **Statement.** Across matched pairs, the order-induced difference in forgetting of k, ΔF, has a positive Spearman correlation with the order-induced difference $\Delta(\Delta c_e^{D*})$.

- **Decision (descriptive labels).** Supported if the lower bound of the 95% bootstrap interval (resampling unit: the matched pair) is > 0.20; equivalent if the interval lies within (−0.20, 0.20); not part of the confirmatory family.
- **Comparator.** The same correlation is reported for C5b (Σ η‖g̃‖), which is also order-sensitive. H3 shows that c_e tracks a path effect invisible to order-invariant counts; it does not, by itself, show an advantage over order-sensitive telemetry.

10.4 Exploratory, labelled as such

D comparisons beyond D*; the SGD arm; T1-acc and T2; C7b, C8, C8b and C9 bounds; single-competitor comparisons; B_standard comparisons; scale-drift robustness checks; per-family results other than those named above.

11. Gates

Gate	Check	Criterion	Consequence of failure
G1, ceiling	Within-run, episode-weighted correlation r_AB between F_A and F_B (weights of §6), uncorrected, on calibration runs	$\bar{r}_{AB} \geq 0.70$ on F2–F3	Increase probe-set size before freeze; report all skills also as a fraction of $\bar{r}_{AB}$
G2, identifiability	Out-of-sample R^2 for predicting the interval rate $\bar{\rho} = \Delta c_e^{D*} / \max(\Delta N, 1)$ from B_strong, with OLS and with GBDT: at Stage 0 by cross-fitting on the F2–F3 calibration runs (§12a); at Stage A fitted on training runs and evaluated on test triplets of F2, F3, F5, F6 and of F5 alone	Both $R^2 < 0.90$	H1 is reported as *not identifiable*, never as equivalence
G2b, H2 denominator	Mean ΔR^2 of $Disp^{D*}$ over B_strong on the H1 test runs (F2 and F3)	Lower bound of the two-sided 95% BCa interval > 0.02	H2 is reported as *not identifiable*
G3, competence and headroom	On calibration runs, out of fold (§12a), verified on validation runs: B_strong beats X_cov alone in F1; headroom = $\bar{r}_{AB} - \bar{R}^2$(B_strong) on F2–F3	$\Delta R^2 \geq 0.10$; headroom ≥ 2.5 δ	Redesign the target or the stream before freeze
G3b, baseline saturation	On calibration runs, out of fold: B_strong re-tuned with a doubled search (100 configurations instead of 50), same seeds for the first 50	Change in mean R^2(B_strong) ≤ 0.01	The baseline is under-tuned: increase the search budget for every predictor set before freeze; if not achievable, H1 may not be reported as supported
G4, leakage and placebo	R ∩ probes = ∅, and V_target, V_ref and V_distractor pairwise disjoint (asserted in code); incremental value over B_strong of a **block-permuted** c_e: per-step increments are cut into 256-step blocks (idle zeros kept in place), the blocks are permuted **across the runs of the same family**, so that no run keeps its own mean rate, the index is re-accumulated per run, and the model is **refitted** on training runs with locked hyperparameters and evaluated on test runs. Checked at Stage 0 (out of fold) and at Stage A, within each family	Point estimate ≤ 0.005 and upper bound of the one-sided 95% interval < 0.015	Stop; leakage or a design artefact

Gate	Check	Criterion	Consequence of failure
G4b, positive control (sensitivity)	On calibration runs only, out of fold: a synthetic predictor Z = e + ξ is added to B_strong, with e the out-of-fold residual of T1 given B_strong and ξ independent Gaussian noise of per-run variance $\sigma^2_{\xi,r} = s^2_{e,r}[(1 - R^2_r(B_{strong})) / (2\delta) - 1]$, $s^2_{e,r}$ the episode-weighted residual variance, so that the expected within-run ΔR^2 is 2δ. Z has monotonic_cst = +1. The full pipeline (weighting, locked hyperparameters, GBDT, bootstrap, decision rule) is applied	Estimated ΔR^2 in [1.25δ, 2.5δ], and outcome *supported*	The pipeline cannot detect an effect of the size sought: fix the pipeline before freeze. If it still fails, an *equivalent* outcome of H1 may not be reported as closing Level 2
G5, local statistics	$\bar{R}^2$(X_cov + c_e) versus $\bar{R}^2$(X_cov + C6) on the H1 test runs, both models tuned with the same 50-configuration search	Lower bound of the one-sided 95% interval of the difference > 0	c_e is reported as failing admissibility criterion (a), whatever H1 shows
G6, onset versus skill	H1 statistic with the number of training runs halved and doubled; Stage B deployment models fitted on the doubled Stage A training set (60 runs)	No flip between *supported* and *equivalent*, and |change in mean ΔR^2| ≤ 0.01	Report as onset-dependent
G7, reproducibility and budget	Bit-identical reruns of a random 5% of runs; configuration hashes verified before analysis; realised rehearsal steps = Q on every Stage B run, with the terminal flush logged	100%	No analysis until fixed
G8, reliance	For S-c_e and S-pred(B_strong + c_e): paired difference T3(live) − T3(placebo twin) over Stage B seeds (§9)	Lower bound of the one-sided 95% interval > 0, and point estimate ≥ δ_B / 2	c_e is decodable but not used
G9, H3 identifiability	Split-half reliability (A vs B) of ΔF across matched pairs; cross-seed reliability of Δ(Δc_e^{D*}) between the two seeds of each saved state; variation SD(Δ(Δc_e^{D*})) / mean of the branch-average Δc_e^{D*}	≥ 0.50; ≥ 0.50; ≥ 0.05	H3 is reported as *not identifiable*

12. Data splits and sample size

- **Calibration runs:** 24 (8 per F1–F3), used only in Stage 0 (§12a): G1, G2, G3, G3b, G4, G4b, the SESOI value, D*, m_ch, the stationarity check of R, the moment-reset diagnostic, the variance of the test statistics, and threshold design. All model-based quantities on them are out of fold (§12a). They are discarded afterwards and never enter test statistics.
- **Training runs:** F1–F3, 30 base. For G6 the models are refitted on 15 of them and on 60 (the 30 base runs plus 30 additional runs), so there are 60 training runs in total. **Validation runs:** F1–F3, 15. **Test runs:** new seeds of F1, F2, F3, F4 (sanity), F5 and F6. O blocks are used only for H3 and G9 (30 saved states × 2 seeds).
- **Sample size for H1** (F2 and F3). Let $\hat{\sigma}_{up}$ be the upper 80% chi-square confidence bound of the SD of per-run ΔR^2 on the F2–F3 calibration runs (out of fold). For TOST at a true ΔR^2 of 0 and for the MET at a true ΔR^2 of 2δ (an effective shift of δ from the boundary δ), each at power 0.80 and one-sided α = 0.025 (for TOST the type II error is split between the two sides, hence $z_{0.90}$):
  - $n_{TOST} = \lceil (z_{0.975} + z_{0.90})^2 (\hat{\sigma}_{up}/\delta)^2 \rceil = \lceil 10.51 (\hat{\sigma}_{up}/\delta)^2 \rceil$;
  - $n_{MET} = \lceil (z_{0.975} + z_{0.80})^2 (\hat{\sigma}_{up}/\delta)^2 \rceil = \lceil 7.85 (\hat{\sigma}_{up}/\delta)^2 \rceil$;

    - n_H1 = max(n_TOST, n_MET), split equally between F2 and F3.
- **Family checks.** n_F5 = n_F6 = n_MET, which gives the F5 replication test (lower bound > 0) and the F6 check (upper bound < 0) power 0.80 at a true effect of ±δ. n_F1 = 10 and n_F4 = 5, reported descriptively.
- **Total.** N_test = n_H1 + n_F5 + n_F6 + n_F1 + n_F4 ≤ 200 (budget cap). If the cap is exceeded, the scope is reduced before freeze, first by dropping F6, then F5 (whose replication check is then reported as underpowered), and the reduction is declared in the Stage 0 report. If n_H1 alone exceeds 200 − n_F1 − n_F4, H1 cannot be run at the prespecified SESOI: the study stops at Stage 0 and the reason is reported (§12a). The SESOI rule is not relaxed.
- **Sample size for H2.** With $\hat{\sigma}_{up}(Z)$, the upper bound of the SD of Z_r on calibration runs, and the margin $\delta_{H2} = \max(0.01, 0.20 \cdot \Delta\bar{R}^2_{cal}(Disp^{\{D^*\}}))$: $n_{H2} = \lceil (2.394 + z_{0.80})^2 (\hat{\sigma}_{up}(Z)/\delta_{H2})^2 \rceil = \lceil 10.47 (\hat{\sigma}_{up}(Z)/\delta_{H2})^2 \rceil$ (non-inferiority at the first Holm level, one-sided $\alpha = 0.025/3$, true $Z = \delta_{H2}$). If n_H2 > n_H1, H2 is reported as underpowered.
- **Sample size for H4a, H4b.** With δ_B and the upper bound of the SD of paired Stage B differences, the larger of TOST and MET at the first Holm level: $n = \lceil (2.394 + z_{0.90})^2 (\hat{\sigma}_{up}/\delta_B)^2 \rceil = \lceil 13.5 (\hat{\sigma}_{up}/\delta_B)^2 \rceil$ (the MET alone would need 10.47). Stage B uses n_B = max(40, n) paired seeds per scheduler, frozen after the power simulation; if the compute budget cannot accommodate n, H4a and H4b are reported as underpowered.
- **Power simulation.** Before freeze, these closed-form sizes are checked by simulation on calibration data, with the weighted R^2, the stratified bootstrap with training-run replicates, the fixed-sequence and Holm procedure, and the separate family checks.
- **Fixed sizes (indicative; frozen after the power simulation):** calibration 24 runs (8 per F1–F3); training 60 (30 base and 30 more for G6); validation 15; O blocks 30 saved states × 2 seeds; Stage B max(40, n_H4) paired seeds per scheduler.

12a. Stage 0: calibration and go/no-go decision

Stage 0 uses 8–12 technical runs, followed by the 24 calibration runs of §12. It measures and decides; it tests no hypothesis. The author decides after Stage 0 whether to continue. Continuation requires all of the following; otherwise the design is simplified again or the study stops, and the decision is reported.

Out-of-fold scheme. All model-based quantities on calibration runs (G2, G3, G3b, G4, G4b, the SESOI value, $\hat{\sigma}_{up}$, $\hat{\sigma}_{up}(Z)$, and the choice of D*) are computed by 4-fold cross-fitting at the run level, stratified by family (18 runs for fitting and 6 for evaluation in each fold), with hyperparameters chosen by an inner 3-fold cross-validation on the fitting runs.

Criterion	Requirement
Target measurable	G1 passed ($\bar{r}_{AB} \geq 0.70$ on F2–F3); at least 20 valid exposure episodes per run; forgetting with non-negligible variance
Index not reconstructible	G2 passed on the F2–F3 calibration runs (interval rate, OLS and GBDT)
Headroom	G3 passed (headroom ≥ 2.5 δ)
Baseline and pipeline	G3b passed (baseline saturated); G4 passed (no leakage; placebo null); G4b passed (positive control detected)
Index and battery	D* and m_ch selected and frozen; stationarity of R passed, or R split and the calibration runs repeated (§5.1); G5 and G2b recorded for D*

Criterion	Requirement
Cost	Measured time per full run (training, probes, D3 and C10b, snapshots for D*) ≤ 25 min single-threaded with eight workers running concurrently; projected Stage A (calibration, training, validation, test and G7 runs, plus model fitting) ≤ 150 CPU-hours; runs resumable from checkpoints; parallel runs bit-identical (G7)
Sample size	N_test (§12) ≤ 200 for the declared set of families

The feasibility of the full design is not assumed: the step times in §14 cover training only, and probe evaluation, D3 and C10b are measured in Stage 0.

Stage 0 report. The values frozen at Stage 0 (δ, D*, m_ch, $\hat{\sigma}$_up, the sample size per family, and the decision on R) and all gate results are written to a report that is deposited with the preregistration (§17).

13. Stage C: transfer to pretrained language models

Stage C keeps the **question and the inferential logic** (B_strong versus B_strong + c_e, three-way outcomes, gates). It does **not** keep every operational definition, and it is **not part of this preregistration**. A separate Stage C adaptation document will be frozen and deposited with its own DOI before any Stage C run. Its cost is not included in §14. Changes already known:

- **Agent.** Publicly available pretrained causal LMs (the Pythia family); a CPU feasibility check on the smallest model; the confirmatory model size and optimizer arm are named in the adaptation document before any Stage C run.
- **No childhood phase.** Pretraining replaces it; R is a fixed generic-text battery evaluated, not trained.
- **Targets.** T1 is the loss on held-out samples of earlier domains, read at fixed target positions. The accuracy-based inclusion criterion is replaced by a loss-based learned criterion defined in the adaptation document.
- **Covariates.** Task type is replaced by domain identity; competence uses a held-out half of each domain's samples; C7a only (no oracle labels).
- **LoRA.** D is computed on the effective weight update $\Delta W_{eff} = (\alpha/r)(B_{\tau+1}A_{\tau+1} - B_\tau A_\tau)$, not on the concatenated adapter matrices.
- **D3.** Restricted to a small number of target positions and to top-K or sketched distributions; the vocabulary of ≈ 50,000 tokens makes full distributions infeasible.
- **Displacement and C8b, C9.** Evaluated only for the smallest model, for LoRA, or for a subset of tasks, as offline references.
- **Task identity.** Stage A marks each task with a task token (task-incremental learning). Text domains in Stage C carry no such token (domain-incremental learning), and part of the measured forgetting can then be a misalignment of the output layer rather than a loss of knowledge (spurious forgetting; Zheng et al. 2025). The adaptation document states whether a domain token is given and reports a readout-realignment check.

The adaptation document must fix, before any Stage C run:

1. **Decision unit.** One primary confirmatory arm: a single named Pythia checkpoint, chosen by a timing check before any forgetting outcome is seen, adapted with LoRA and AdamW (the regime of FOREVER). Full fine-tuning of the smallest feasible model is secondary. If both are run and their three-way outcomes differ, Stage C is reported as regime-conditional, not as reproduction.
2. **Resampling unit.** A Stage C run is an independent seed controlling the draw and order of domain segments from a pool of at least 15 domains, disjoint document subsamples for training and for probe halves A and B, the stream schedule, and the initialisation of the adapters.

3. **LoRA distances.** D1 and D1_rel on ΔW_eff, computed through the rank-2r factorisation $\|\Delta W_{eff}\|_F^2 = (\alpha/r)^2\, Tr(\tilde{B}^T\tilde{B}\,\tilde{A}\tilde{A}^T)$, *with* $\tilde{B} = [B_{\tau+1}, -B_\tau]$ and $\tilde{A} = [A_{\tau+1}; A_\tau]$, without forming dense matrices; D2 and C5d in the adapter coordinates, where the optimizer moments exist.
4. **Top-K D3.** √JS on the union of the top-K tokens (K = 128) at both times plus one tail bucket, so that both distributions share a support and √JS remains a metric.
5. **Inclusion and covariates.** A loss-based inclusion criterion: a relative reduction of half-B loss over the exposure segment of at least γ_inc, calibrated so that $\bar{r}_{AB} \geq 0.70$; input overlap over a fixed histogram of the 4,096 most frequent tokens plus a tail bin.

The Stage A conclusions are claimed to generalise only if the primary Stage C arm reproduces the three-way outcome of H1.

14. Compute estimate

Measured at drafting on the author's laptop (16 logical CPUs), with the §3 agent (0.93 M parameters, untied embeddings, AdamW, clipping):

Setting	ms per training step
batch 32 × 4 tokens, 1 thread	19.9
batch 32 × 4 tokens, 8 threads	17.3
batch 32 × 64 tokens, 1 thread	206.5
batch 32 × 64 tokens, 8 threads	87.5

Because examples are 4–6 tokens long and unpacked (§3), the relevant figure is ≈ 20 ms per step on one thread; the shorter maximum context of v0.3 does not change it. The cost of eight concurrent single-threaded workers (memory bandwidth, thermal limits) is measured in Stage 0 (§12a).

Item	Estimate (to be replaced by Stage 0 measurements)
Training, ≈ 23,000 steps	≈ 8 min per run
Probe evaluation (end of every segment and every 256 steps; ≈ 130 evaluations × up to 40 tasks × 256 items, plus R and B_beh; batched, inference mode)	≈ 2–3 min per run
D3 and C10b every 8 steps (≈ 2,900 chunks × 256 items, one forward pass for both)	≈ 2–3 min per run
Snapshots for D* (worst case 40 × 1.9 M floats)	≤ 0.3 GB per worker
One run	≈ 12–15 min single-threaded, to be measured under eight concurrent workers
Stage A (24 calibration + 60 training, including G6 + 15 validation + ≤ 200 test + ≈ 16 G7 reruns ≈ 315 runs, plus model fitting with the bootstrap replicates)	≈ 65–85 CPU-hours, ≈ 9–11 wall-clock hours on 8 workers
Stage B (validation: ≈ 10 threshold schedulers × 15 seeds; test: ≈ 7 schedulers, 2 placebo twins and 2 yoked arms × 40 seeds; ≈ 590 runs)	≈ 120–150 CPU-hours, ≈ 15–19 wall-clock hours on 8 workers
C8b and C9 (exploratory)	on a subsample of 30 test runs
Stage C-mini (smallest Pythia model, CPU)	hours per run; feasibility only

15. Interpretation table (frozen with the protocol)

"Gates passed" means G2, G3b, G4 and G5 passed. Rows on H1 are mutually exclusive.

Outcome	Interpretation
H1 supported, gates passed, F5 replication passed, F6 upper bound ≥ 0	Accumulated own change predicts own forgetting beyond clocks, counts, and a strong set of logged summaries; the result generalises across generator parameters and is not misleading under change without interference: first empirical support for the experiential index (Level 2)
H1 supported, gates passed, F5 replication passed, F6 upper bound < 0	The index generalises across generator parameters and is useful where change and interference coincide, but misleading where they decouple; reported as a conditional result
H1 supported, gates passed, F5 replication not passed	The advantage does not generalise beyond the training generators; reported as a construction effect, not a general result
H1 lower bound > δ, but G3b failed	Not interpretable as support: the baseline was not saturated
H1 lower bound > δ, but locked and re-tuned outcomes diverge (§8)	Inconclusive: sensitive to hyperparameters
H1 equivalent, and G3b and G4b passed	Counts and logged summaries suffice; Level 2 is closed for this regime
H1 equivalent, but G4b failed	Not interpretable as closure: the pipeline could not detect an effect of the size sought
H1 inconclusive	Level 2 is unresolved at the prespecified sample size and SESOI; neither support nor closure is claimed, and Stage B is not run
c_e beats B_standard but H1 is equivalent	The information in c_e is reconstructible from cheap telemetry
G2 fails	Not identifiable in this design
G4 fails	Stop: leakage or a design artefact; no result is reported as a finding
G5 fails	c_e fails admissibility criterion (a), whatever H1 shows
G6 fails	The result is reported as onset-dependent
F1 alone	Equivalence on F1 is the expected result of the positive control and does not weaken H1
H2 supported	The scalar index captures at least 80% of what full displacement captures, at the storage ratio reported for D*
H2 refuted	Displacement carries information that the path index does not; the claim of a compact index is weakened
H2 inconclusive, or G2b fails	Neither a retention claim nor its rejection is supported
H3 supported (G9 passed; optional module)	The index tracks a global path effect invisible to order-invariant counts (exploratory)
H4a supported, G8 passed, task-selection check passed	The agent uses the index inside a covariate-aware predictor to improve a decision about itself
H4a supported, task-selection check failed	Reported as a duplicate of the timing test, not as evidence about task selection
H4b supported and G8 passed	The index alone suffices as a rehearsal trigger, beyond the best cheap trigger chosen on validation
H4a or H4b supported, but G8 failed	The gain does not depend on the temporal alignment of c_e: c_e is decodable but not used, and criterion (d) is not met

Outcome	Interpretation
The primary Stage C arm reproduces H1	The result is not an artefact of the synthetic world

16. Threats to validity and mitigations

Threat	Mitigation
Designer degrees of freedom in the stream families	F1 natural; F5 and F6 held out; F6 adversarial; families frozen; Stage C on real data
The mixture of families decides H1	H1 confirmatory on F2 and F3 only; F1, F5 and F6 reported alone with prespecified checks (§10.2)
Omitted cheap competitor reconstructs c_e	B_strong with LR-weighted, momentum (weighted and unweighted), work, weight-decay path, norm, local and canary summaries (signed and accumulated); c_e and competitors on the same footing (raw and rate); G2 on the rate; B_standard reported; B_strong^{+} as sensitivity
Shared noise between predictors, inclusion, and target	Half B for covariates and inclusion, half A for the target; R and the three key vocabularies disjoint from probes and from each other; block-permuted placebo G4; T2 not confirmatory
Scale and weight-decay drift	D1_rel; C5f; C6b; $N(t_0)$ and state norms in X_cov; T1-acc
Deterministic drift of unused parameters	Maximum context 8; no weight decay on biases and LayerNorm parameters
Local-statistics shortcut (closed study)	C6 with short clamped windows inside B_strong; G5
Over-weighting of early tasks	Equal weight per exposure episode
Dependence of triplets within a run	Run-level inference; within-run weighted R^2
Training-sample variance ignored	250 training-run bootstrap replicates for both model families (§8)
Selection of D, m_ch or comparators after seeing results	D*, m_ch and S-C* chosen on calibration or validation and frozen
Hyperparameter locking hides an interaction	Divergence rule (§8); unconstrained sensitivity fit
Unequal rehearsal budgets and pacing artefacts	Hard cap; corrected pacing rule; θ_cal calibrated for uniform spending; terminal flush ≤ 5% and logged; G7
Train–deployment shift for S-pred (probe covariates offline, online covariates and closed-loop intervals in Stage B)	Separate Stage B deployment model on online covariates (§8); in-loop calibration logged; task-selection check (§9)
Under-tuned baseline produces a spurious advantage	Equal tuning budget; locked hyperparameters; saturation gate G3b
A null result produced by an insensitive pipeline	Positive control G4b
Optimizer specificity	SGD arm (exploratory)
Scale	Stage C; restrained claims

17. Deliverables

- Code (MIT), data and reports (CC BY 4.0), frozen configurations with SHA-256, a run inventory, automated audit checks, and scripts that regenerate every reported table.

- A Zenodo preregistration (with DOI), deposited after Stage 0 and before any Stage A or B training, validation, or test run. It states explicitly that the study is a change of the plan announced in the Perspective (versions 1.0–1.2) and contains the frozen protocol, the Stage 0 report (§12a), the SHA-256 hashes of the locked values (D*, m_ch, δ, σ̂_up, sample sizes per family) and the test-run seed lists, or a stopping report if Stage 0 halts the study.
- A preprint with *References* links to the Perspective and the closed study, and *Is supplemented by* to the code archive.

18. References

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All references were checked against arXiv, Crossref or the publisher; those added in v0.2 and v0.3 on 3 November 2026.

19. Revision log v0.1 → v0.2

Sources: **A** = Google Argon review of v0.1 (truncated in Q12; a continuation was requested); **C** = Copilot analysis of that review; **O** = the author's own pre-review. Every accepted item was checked before adoption.

Accepted

Change	Sections	Source
Half B for covariates and inclusion; half A for the target; fact tasks enlarged to 256 items; balanced exposure	§4.1, §4.2, §6, §7	A, O
G1 ceiling uses the uncorrected r_AB (the target uses one half)	§7, §11	A, O
Same-step D2 cancels the Adam preconditioner (verified algebraically); D2 uses a segment-lagged $\hat{v}$; C5d added; Disp^{D2} anchored at $\hat{v}(t_0)$	§5, §7	A, O
D1_rel against scale and weight-decay drift; $\Delta\theta$ logged before and after decay	§3, §5	A
D2 defined in the SGD arm by a passive EMA	§3	A, O
Clipped and raw gradient norms both logged and used	§3, §7	A
B_strong with LR-weighted gradient and loss, squared gradient, SI work, momentum norm, noise scale, state norms, $N(t_0)$; B_standard kept descriptively	§7	A (two-tier design: C)
Canary loss C10 on R	§7	O
C6 windows shortened to $W_{eff} = \min(20, \lfloor \Delta N/4 \rfloor)$ and clamped; minimum interval of 50 steps	§6, §7	A
π_overlap removed from X_cov; C7 split into agent-observable C7a and oracle C7b; agent-observable novelty and overlap covariates	§7	A
R and B_beh on a disjoint key pool; stationarity check of D3	§4.1, §5	A
Storage accounting: one running scalar plus one snapshot per task; snapshot objects per D; C9 storage and training-items gradient	§1, §5.2, §7	A
T1-acc secondary target	§6	A
Operational specifications: answer-position reading, base-2 JS, vocabulary partition, rule-task template, no packing, fixed childhood with gate, O blocks forked from one state, fixed C1 basis, end of interval before the next exposure	§3–§7	A
Monotone constraints specified per feature; locked-hyperparameter ΔR^2; ridge transforms and scaling	§8	A

Change	Sections	Source
Equal weight per exposure episode	§6, §8	A
Cross-fitting over training runs; two-sample bootstrap for ridge only	§8	A (scope: C)
H2 as the linear contrast Z_r with denominator gate G2b and an equivalence bound	§10, §11	A, O
H3 redesigned around a global property and moved outside the confirmatory family; G9 expanded; threshold 0.20	§10.3, §11	A (secondary status: C)
Stage B: hard budget cap, pacing, terminal flush; comparator chosen on validation; S-uniform, S-C2, S-C5, S-C5b added; H4 split into H4a and H4b; online covariates from training batches	§9, §10	A, O
Within-run block-permuted placebo; G4 one-sided, locked hyperparameters, per family; cross-seed yoked stream kept as a robustness check	§9, §11	A (both kept: C)
Single $\alpha = 0.025$ per side for MET and TOST; fixed sequence H1, then Holm over H2, H4a, H4b; Stage B only after H1 is supported	§9, §10	A, C, O
Power formulas with $z_{1-\beta/2}$ for TOST, the Holm level, an upper bound on σ, and F5/F6 allocation; checked by simulation	§12	A, C
Headroom-based SESOI rule, frozen before test, with raw and relative gains both reported	§10.1	A (caution: C)
G2 on the interval rate with OLS and GBDT; G3 on calibration with a headroom check; G5 and G6 made quantitative; G8 one-sided	§11	A, O
F4 not pooled; adversarial F6 added; F5 described as generalisation across generator parameters; F1 row added to the interpretation table	§4.4, §15	A, O (F6 design: C)
Stage C rewritten as a transfer protocol with a separate adaptation document; LoRA on ΔW_{eff}; no childhood	§13	A, C, O
Compute estimate replaced by measured step times; single-threaded workers for determinism and throughput	§3, §14	O, A
Notation: $\bar{\rho}$ for the rate, m_{ch} for the chunk stride, L_{reh} for the rehearsal length; §2 wording on elapsed time; consistent scheduler names	throughout	A
Content versus provenance note	§0.4	O
Prior work FOREVER (Feng et al. 2026) cited, with priority for the index acknowledged; contribution restated as an identification audit; exploratory S-FOREVER arm. Verified on arXiv and in the ACL Anthology: $\tau_t = \Sigma\|\Theta_i - \Theta_{i-1}\|_2$ on LoRA weights; only step counts tested as an alternative clock	§0.3, §9, §18	A (named), C (positioning), O (verification)
H3 made an optional exploratory module, run only if H1 is supported and budget remains	§10.3, §15	C
Stage 0 go/no-go criteria (cost, sample size, episodes per run)	§12a	C
Stage C declared outside this preregistration	§13	C
Positive control G4b (sensitivity of the pipeline to an effect of 2δ) and baseline-saturation gate G3b (doubled tuning budget); interpretation table and Stage 0 criteria updated. Motivated by the second priority-search round on the closed study (positive-control validation; Mukhoti et al. 2018)	§11, §12a, §15	C, O
Mukhoti et al. 2018 (strong baselines), Knopov and Shvai 2026 (TOST and run counts for stochastic ML), Elazar et al. 2021 (decodability is not use) cited; all verified on arXiv or Crossref	§7, §9, §10.1, §18	C (priority search on the closed study)

Rejected or modified

Suggestion	Decision
"PyTorch 2.13 and NumPy 2.5 do not exist" (A)	Rejected: both are installed and were used for the measurements (PyTorch 2.13.0, NumPy 2.5.1)
Per-step D2 weighted by the running maximum of $\hat{v}$ (A)	Modified: segment-lagged $\hat{v}$ is used, because a running maximum never decreases and would track history age
Weight-decay pulses and pre-LN rescaling as part of F6 (A)	Moved to exploratory robustness checks: no counterpart in ordinary training (C)
H3 option B (midpoint evaluation or a recency-weighted index) (A)	Rejected: it introduces a second index and complicates interpretation (C)
Packed 64-token sequences with block-diagonal masks (A)	Rejected: examples are unpacked, which is simpler and, as measured, ≈ 4× cheaper per step
Argon's split sizes adopted as final (A)	Modified: indicative only, to be fixed by the power simulation (C)
Two-sample bootstrap with refitting for all models (A)	Modified: ridge only; cross-fitting for GBDT (C)
Go/no-go on $R^2(\Delta c_e \mid B_strong) < 0.90$ for the raw increment as well as the rate (C)	Rate only: the raw increment is dominated by interval length and would fail the check trivially
Full-run time of 45–120 min (C)	Not adopted as an estimate: it assumed long sequences; examples are unpacked. Replaced by a measured go/no-go limit of 40 min per run (§12a)

Closed. The truncated part of the review of v0.1 was not continued; v0.2 was reviewed in full instead (§20).

Update of 1 November 2026 (no change of design). The Perspective (v1.3, 10.5281/zenodo.23847648), the study (v1.3, 10.5281/zenodo.23847602), and the code archive (v1.3, 10.5281/zenodo.23847723) were revised after an automated review by the CMU Paper Reviewer. Citations and the closure rule in §0 now follow v1.3: non-inferiority within a prespecified margin and no worse declared cost, with mixed trade-offs reported as unresolved.

20. Revision log v0.2 → v0.3

Sources: **R2** = Google Deep Review (Argon, Medium) of v0.2, 3 November 2026, each point checked against the text before adoption; **O** = the author's own check, including four points of the v0.1 review that v0.2 neither adopted nor rejected; **M** = Mistral Large 4 review of v0.3 (API, reasoning effort "low"), 4 November 2026, each point checked against the text (its check of this log found every accepted change implemented).

Accepted

Change	Sections	Source
H1 confirmatory on F2 and F3 only; F1, F5 and F6 reported alone with prespecified checks; total test runs as a sum of family allocations; $\hat{\sigma}_up$ on F2–F3; bootstrap stratified by family	§4.4, §10.2, §12	R2
Pacing rule corrected (it lowered the threshold when spending ran ahead, a runaway loop); ε fixed; θ_cal calibrated for uniform spending; terminal flush bounded and logged; decisions on the 64-step grid; S-pred triggered by the increase in predicted forgetting since the last rehearsal	§9, §11	R2
c_e and Disp enter as {raw increment, rate}, as the competitors do; +1 on all four terms of C1; matched ridge basis; constraints per column; unconstrained sensitivity fit	§2, §8	R2

Change	Sections	Source
Separate Stage B deployment model fitted on the online covariates; in-loop calibration; task-selection check (≥ 10% non-least-recently-rehearsed choices)	§8, §9, §15, §16	R2
G4: model refitted; blocks permuted across the runs of a family; checked at Stage 0 and Stage A	§11	R2
Stage B placebo made causal (the live twin's realised blocks, permuted and fed online; residual for S-pred); G8 as the live-versus-placebo contrast	§9, §11	R2
Interpretation table rebuilt: mutually exclusive rows on H1; rows for H1 inconclusive, G3b, G4, G6 and G8 failures, divergent hyperparameter outcomes, H2 refuted, and the task-selection check	§15	R2
Stage 0: out-of-fold 4-fold cross-fitting on the calibration runs; G4, stationarity of R, m_ch, G5 and G2b added; cost limit of 25 min per run under eight concurrent workers; sample-size criterion aligned with the 200-run cap; Stage 0 report	§12a, §17	R2
Step quantum of 64 steps, so that 256-item epochs at batch sizes 16, 32 and 64 end exactly at segment ends: childhood 3,072 steps, evaluation grid 256, L_reh = 64, minimum interval 64, m_ch ∈ {1, 4, 8, 16, 32, 64} (default 8), placebo blocks of 256 steps	§0.7, §4.1, §4.3, §5.1, §6, §9, §11	R2 (quantum: O)
Maximum context 8 and no weight decay on biases and LayerNorm parameters: unused positional embeddings would otherwise decay deterministically and inject a Σ η signal into D1 and ‖θ‖	§3, §16	R2
Pre-decay update for D1_rel, D2, C5c and C6; ℓ defined (weight and embedding matrices); D2 with a fixed lag of 256 steps instead of segment starts; C5e (Σ ‖m̂‖) and C5f (weight-decay path) added; logged quantities completed	§3, §5.1, §7	R2
Segment boundaries defined by agent-observable events	§4.3, §7	R2
C6 window W_eff = min(16, ⌊ΔN/4⌋), zeros when ΔN = 0, population SD, pre- and post-decay; inclusion rule for idle intervals; idle intervals in multiples of 64 ticks, evaluated at their end and every 256 ticks	§4.3, §4.5, §6, §7	R2
Vocabulary: two-token keys; token-disjoint key vocabularies V_target, V_ref, V_distractor; task tokens allocated, including distractor tokens; F6 batch sizes as in F2; fixed training set of 256 inputs per rule task; |R| = |B_beh| = 256 (128 fact, 128 rule); overlap over input pairs; log number of distinct target values as a covariate	§4.1, §4.4, §7	R2
Chunked distances written out; D4 exploratory, after the final LayerNorm, standard CKA; square root restored in the same-step identity	§5.1	R2
Displacement formulas written out; metric-free Disp^{D1} alongside if D* = D2; snapshots on test runs only for D*; C8b and C9 on 30 test runs; C8b + C9 as a joint reference	§5.2, §7, §14	R2
R split into R_maint and R_eval if maintenance is needed; accumulated unsigned canary change C10b in B_strong	§5.1, §7	R2
G1 on the episode-weighted r_AB; SESOI from F2–F3 calibration runs, out of fold	§7, §10.1, §11	R2
G4b: per-run noise scaling, +1 constraint on Z, tolerance [1.25δ, 2.5δ]	§11	R2
Divergence rule between locked and re-tuned ΔR^2	§8, §15	R2
H2 as a non-inferiority test (supported, refuted, inconclusive); its sample size with σ̂_up(Z) and a scaled margin δ_H2	§10.2, §12	R2
Search space stated; cross-validation grouped by run; 250 training-run bootstrap replicates for both model families (the v0.2 restriction to ridge is withdrawn: the cost is a few CPU-minutes)	§8, §10.1	R2

Change	Sections	Source
G2 at Stage 0 and Stage A; G2b on the two-sided 95% interval on F2 and F3; G5 as a one-sided lower bound > 0 with both models tuned; G6 wording; G7 terminal flush; G9 pairing (30 states × 2 seeds) and its denominator	§10.3, §11	R2
Compute estimate recounted: ≈ 315 Stage A runs, ≈ 590 Stage B runs, 12–15 min per run, batched probes	§14	R2
Stage C adaptation document: decision unit, resampling unit, LoRA distance algebra, top-K D3 with a tail bucket, loss-based inclusion, overlap histogram; task identity and spurious forgetting (Zheng et al. 2025, verified on arXiv)	§13, §18	R2, O
Optimizer moments: initialisation, persistence, bias correction; SGD scaling of D2	§3, §5.1	R2
Storage wording; weak ordering in the identification note; $\bar{\rho}$ with max(ΔN, 1); conditional ranking inside a learned predictor	§0.7, §1, §2	R2
H1 reported by task type and by tercile of ΔN (descriptive)	§6	O (v0.1 review)
Calibration diagnostic: optimizer moments reset at segment boundaries in a forked copy	§5.1	O (v0.1 review)
References: code archive v1.3; five references of v0.2 verified on arXiv; Zheng et al. 2025 added; placeholder note removed	§18	R2, O
Bootstrap mechanics: each draw pairs the resampled test runs with one training-run replicate; jackknife acceleration	§10.1	M
The pair {Δc_e, $\bar{\rho}$} adds one degree of freedom, stated	§8	M
G6 training-run accounting: 60 training runs in total	§12	M
Stop rule if n_{H1} alone exceeds the cap; Stage B size $n_B = \max(40, n_{H4})$	§12	M
TOST power wording: type II error split between the two sides ($z_{0.90}$ for power 0.80)	§12	M (a misreading showed that the wording was unclear)

Rejected or modified

Suggestion	Decision
Further norm- and preconditioner-scaled accumulators in the confirmatory baseline (R2)	Modified: reported as the sensitivity baseline B_{strong}^{+}. The confirmatory baseline already holds C5d, C5e, C5f, state norms and the normalisation of D1_rel; ratios of the candidate's own ingredients would make G2 fail by construction
F5 replication as "lower bound > 0 and point estimate ≥ δ" (R2)	Modified: lower bound > 0 only. Requiring the point estimate to reach δ halves the power when the true effect is δ
Step quantum of 48 steps, or tasks of 200 items with batch sizes {20, 40, 100} (R2)	Modified: a quantum of 64 steps keeps 256 items and the batch sizes
Lower bound of G2b raised to 0.05 (R2)	Not adopted: the H2 margin is scaled instead ($\delta_{H2} = \max(0.01, 0.20 \cdot \Delta\bar{R}^2_{cal}(Disp))$)
D2 anchored at the curvature at the end of childhood (R2, alternative)	Not adopted: a fixed lag of 256 steps keeps the curvature current
Publication venues added to the arXiv references (R2)	Deferred: each venue is added only after it is checked at the publisher

Suggestion	Decision
G4b noise formula "does not give 2δ" (M)	Rejected: with $Z = e + \xi$, the expected within-run ΔR^2 is $(1 - R^2) \cdot s^2/(s^2 + \sigma^2)$, which equals 2δ for the stated σ^2 (checked numerically: $R^2 = 0.5$, $\delta = 0.05$ give $\Delta R^2 = 0.10$). The review assumed that adding Z removes σ^2 from the residual linearly
n_TOST "uses power 0.90" (M)	Rejected: for TOST at a true difference of 0 the type II error is split between the two sides, so $z_{0.90}$ gives power 0.80; the wording was clarified
Ceiling should be $\bar{r}_{AB}^2$ (M)	Rejected: for parallel halves, the largest fraction of the variance of F_A that any predictor of the true score can explain is the reliability r_{AB} itself; r^2_{AB} would understate the ceiling